

2022-Ph-H1-Q14

Section: Our Dynamic Universe

Topic: Measure, Gravitation, Relativity

A satellite moves in a circular orbit. What provides the centripetal force?

Answer: C

(Gravity)

Guidance for Students:

Gravity keeps satellites in orbit — it pulls them inward, providing the centripetal force needed for circular motion.

Revision Tips:

- Satellites are in **free fall** around Earth.
- Centripetal force = gravitational force.
- Use $F = \frac{mv^2}{r} = \frac{GMm}{r^2}$